Entailment and modeling process

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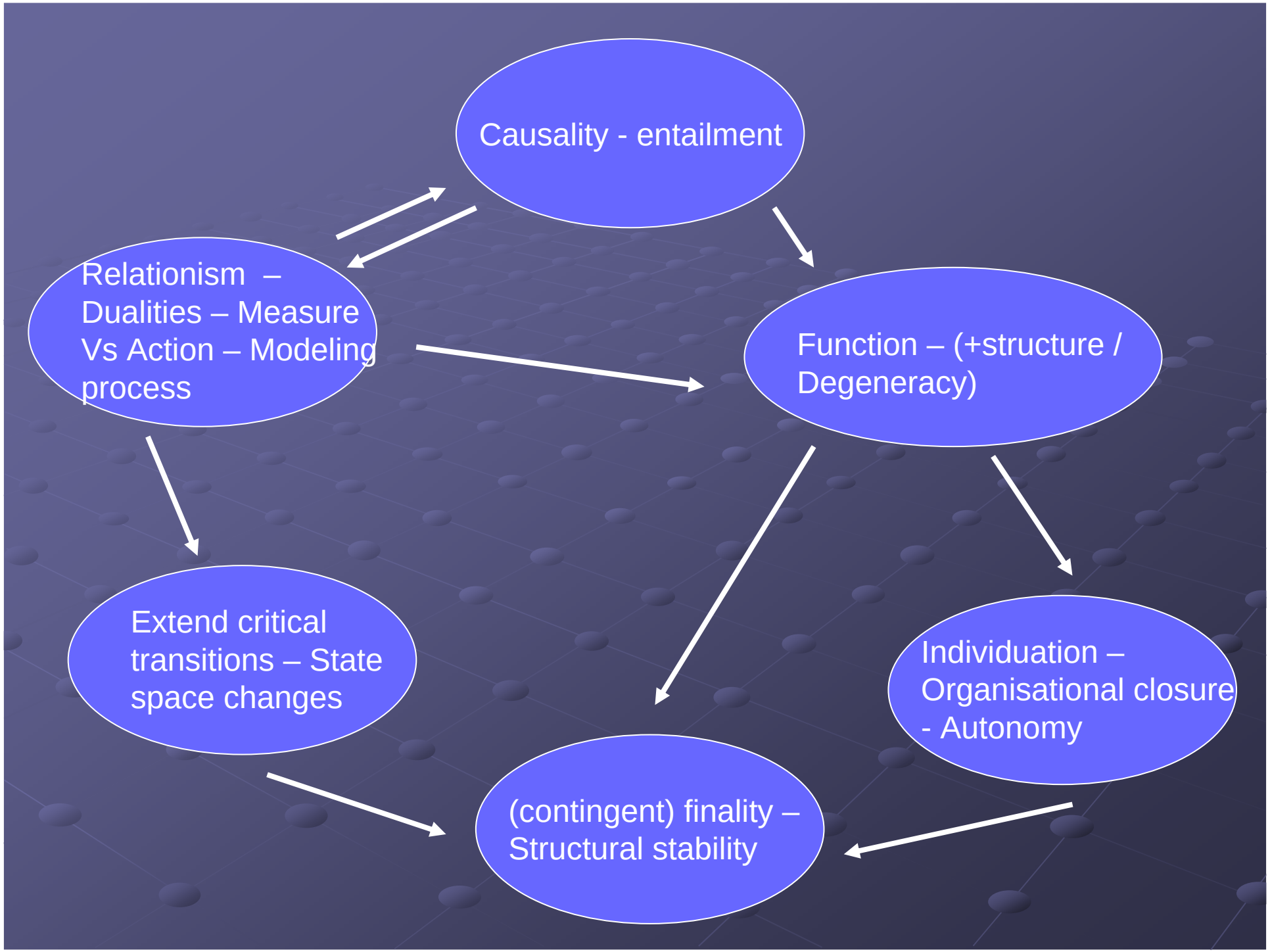
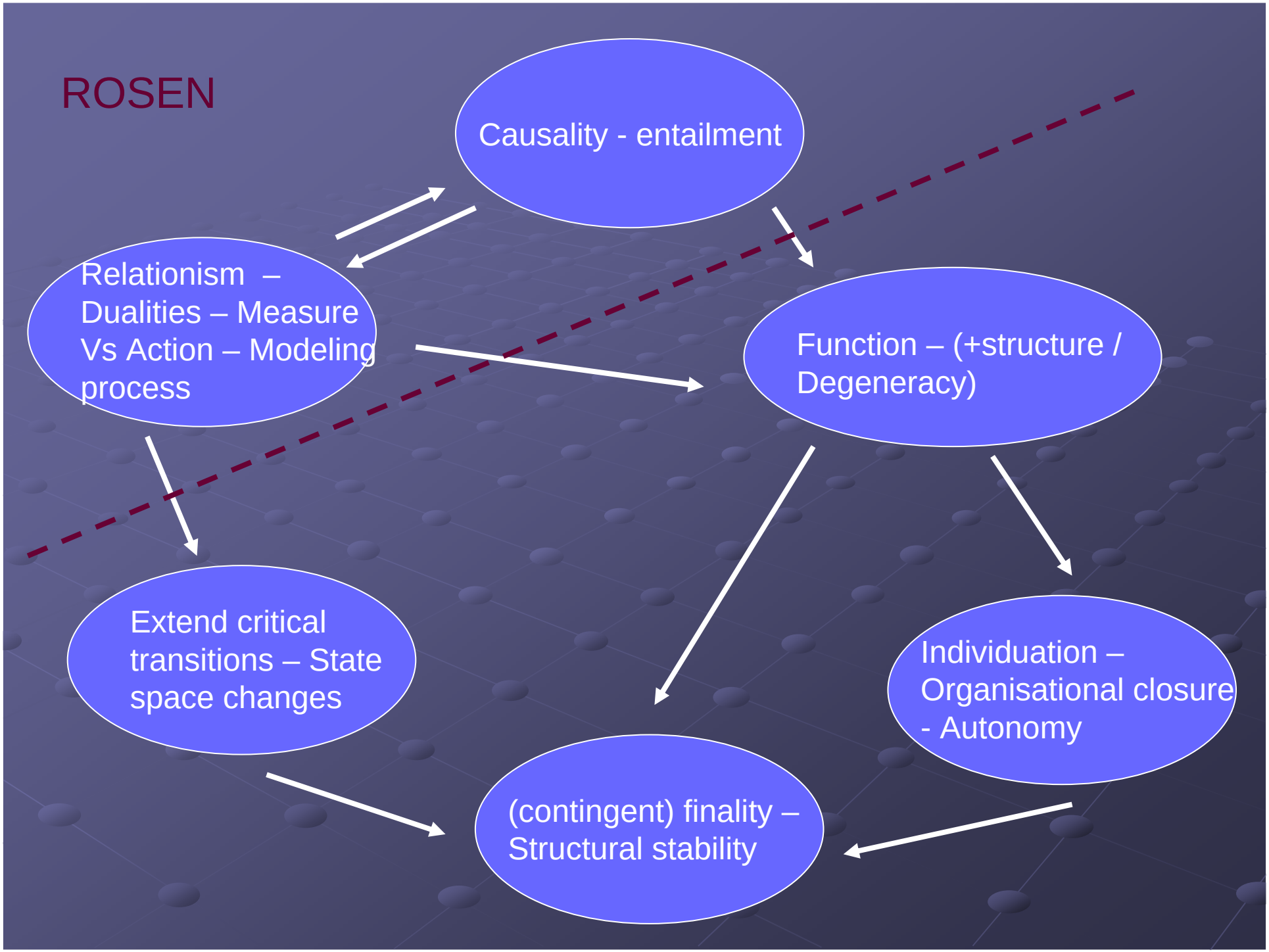

Causality - entailment
Relationism – Dualities – Measure Vs Action – Modeling process
Function – (+structure / Degeneracy)
Extend critical transitions – State space changes
Individuation – Organisational closure - Autonomy
(contingent) finality – Structural stability

ROSEN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- Causality Vs entailment
- Limitations of state-based (Newtonian syntaxic?) mechanics
- Modeling process
- Modeling relation revisited
- Lessons from logic
- Closure under efficient causation

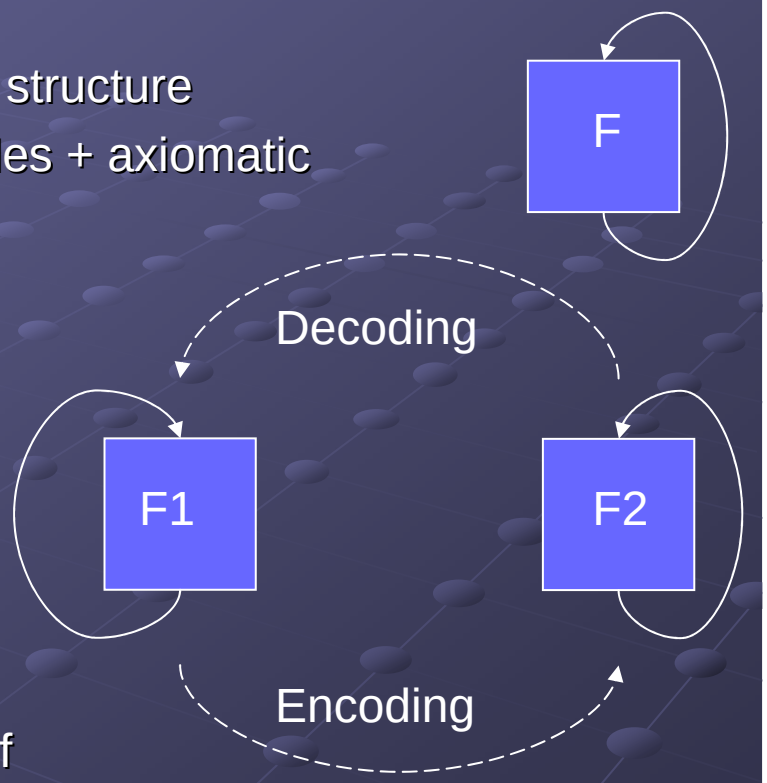
Entailment

Entailment

- « inferential entailment » in language
  - Proposition P entails proposition Q
  - *« Syntactical rules that allows us (without consulting external referents) to say that a group of propositions implies others » (LI, p.43) »*
  - No complete characterization of (natural) language
- Def : formalism
  - *« We shall understand by a **formalism** any sublanguage of a natural language, defined by syntactic qualities alone »* (LI, p.44)
  - *« Finite list of production **rules**, together with a generating family of **propositions** on which they can act »*
- Natural systems Vs Formal systems
  - *« The entire scientific enterprise is an attempt to capturate natural systems within formal ones »* (LI, p.44)
- *« Entailment is (peharps) the central concept in the present work » LI, p.46)*

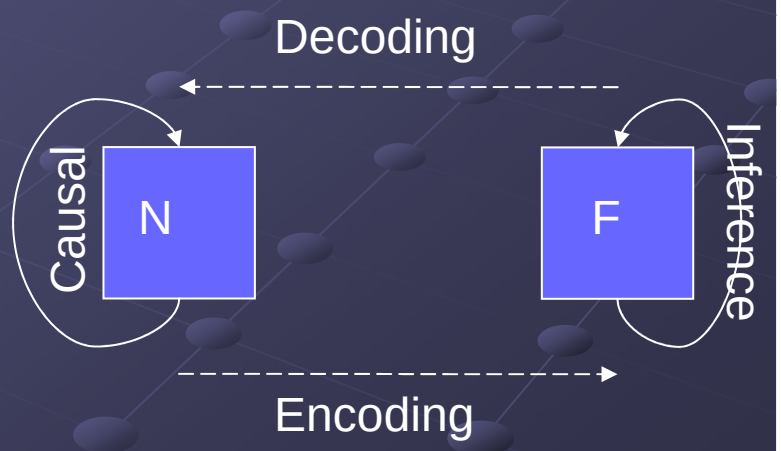
Comparison of formalisms – Definition of a « model »

- Inferential structure :
  - A formalism possesses its own inferential structure
  - Inferential structure = set of production rules + axiomatic propositions
- Comparison of formalisms :
  - Encoding and decoding are (set-theoretic) maps from propositions in one system to propositions in the other
- Def : modeling relation, and **model**
  - F2 is a **model** of F1 (F1 is a **realization** of F2) if the diagram <u>always commutes</u>
  - We will say a **modeling relation** exists between F1 and F2



Causality = entailment between phenomena – Modeling relation

- Is there any kind of entailment in the external world?
  - *« If not we can all go home; science is not only impossible but also inconceivable »* (LI, p.55)
  - What we observe are phenomena, noumena (things themselves) are inherently unknowable
  - « We will suppose that relations of entailments do indeed exist between phenomena (…) this is the province of causality » (LI, p.58)
- Natural law :
  - = **modeling relation** between causal entailment in natural system and syntactic entailment in formal system
  - Asserts :
    - There are relations between phenomena
    - Mind can perceive theses relations



Modeling relation? (1)

- **H. Hertz** (1857-1894), *The Principles of Mechanics*, Dover, NY, 1984, pp.1-2 [orig. German ed., *Prinzipien Mechanik*, 1894]
  - We form for ourselves images or symbols of external objects; and the form which we give them is such that the logically necessary (denknotwendigen) consequents of the images in thought are always the images of the necessary natural (naturnotwendigen) consequents of the thing pictured.

    For our purpose it is not necessary that they [images] should be in conformity with the things in any other respect whatever. As a matter of fact, we do not know, nor have we any means of knowing, whether our conception of things are in conformity with them in any other than this *one* fundamental respect.
- Pattee :
  - We are free to use any kind of symbol system or imagery, finite, infinite, continuous discrete, algorithmic or heuristic, computable or not, crisp, stochastic, or fuzzy logic, and so on. The only test is the commutation: "the consequent of the image in thought (model) conforms to the image of the consequent in nature."

Pattee's ELABORATED HERTZ/ROSEN MODEL COMMUTATION DIAGRAM - See Anticipatory Systems, p.74 or Life Itself, p. 60

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  OBJECT        EPISTEMIC CUT        IMAGE

 INITIAL            |
 STATE (t)___ MEASUREMENT 1___INITIAL CONDITIONS(t)
    |               |                |
 NATURAL            |             INFERENTIAL   |
  CAUSAL                            MODEL (LAWS)
  WORLD                               |
    |                             CONSEQUENT
CONSEQUENT          |             OF IMAGE \
  STATE             |                |      TEST
  (t + s)____ MEASUREMENT 2 _____IMAGE OF /
    |               |             CONSEQUENT
  TIME
   ||
   \/
```

Tim Gwinn – 12/2004 – Importance of the commuting modeling relation

- Rosen's philosophy of science and his arguments all flow from the commuting (in Rosen's sense) modelling relation being the epistemological underpinning. Without the requirement for the modelling relation to commute (in Rosen's sense), the whole epistemology collapses and there is no force of argument for the existence of Rosennean complexity or a requirement for complementary models or analytic <> synthetic or for any importance of organization, a distinction between models and simulations, etc.

Modeling relation? (2)

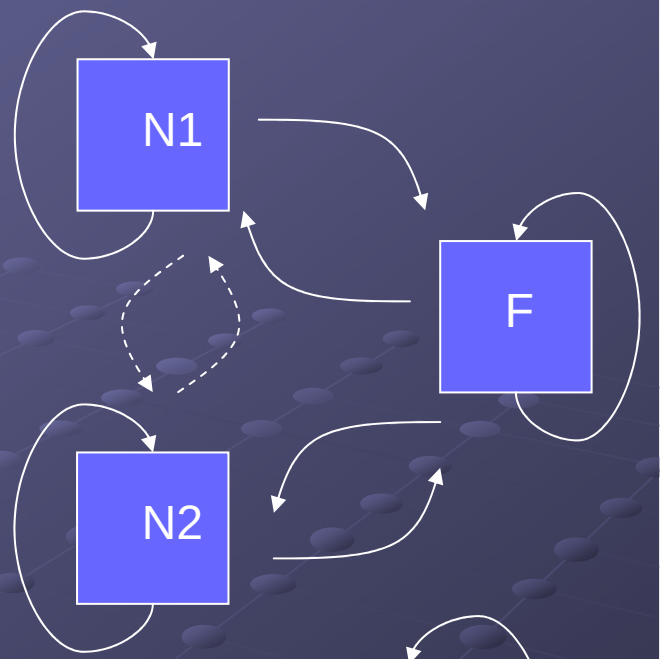
- Helmholtz, H. (1878/1977) E*pistemological writings*, The P. Hertz/M. Schlick Centenary Edition of 1921, English translation by M.F. Lowe, Reidel Publishing Company.
  - "*When we perceive before us the objects distributed in space, this perception is the acknowledgement of a lawlike connection between our movements and the therewith occurring sensations* […]*. What we perceive directly is only this law*" – Helmholtz (1878/1977)

Modeling relation? (3)

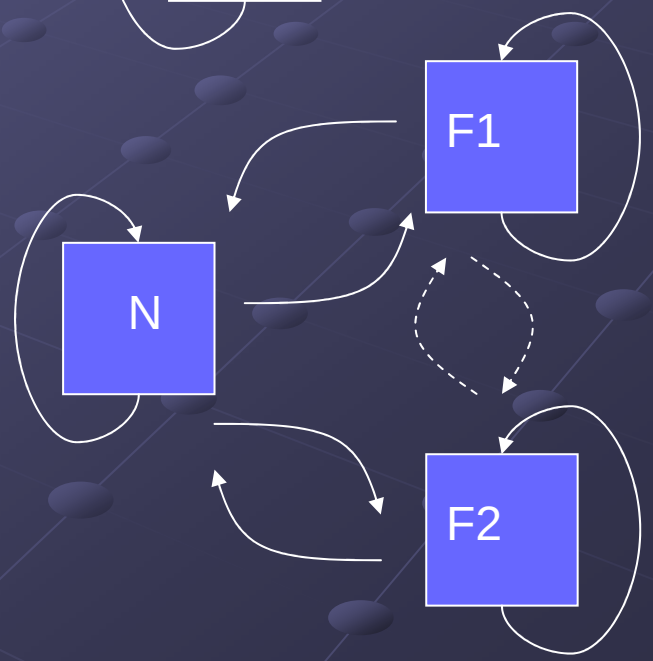
- Palmer, S. E. (1999), *Science Vision: photons to phenomenology,* The MIT Press, Cambridge, Massachusetts.
  - *"A representational system can be analysed as a homomorphism: a mapping from objects in one domain to objects in another domain such that relations among objects in the first domain are mirrored by corresponding relations among corresponding objects in the representation"* – adapted from Palmer (1999).

Modeling relation (4) - Aspects et découpeurs chez M. Mugur-Schächter

- 2 natural systems have a same model



- Natural system N has 2 models



Causality Vs Aition (=ways of answering to a « why? » question)

Av=w (ELI, p.165)

- A is an operator, v an argument, w is a value
- To ask for the causes of w is unnatural
- This is an inferential system, not a causal one
- Speaking of « cause » is metaphorical. And cause is not what we are after. Use of the word cause generates confusion
- Instead we are looking for answers to a « why? » question
- Ask for the *aitia* of w : it means you think in terms of the four fashions (Aristotle)
  - V is the raw material, the material aition of w
  - The operator A imparts a form (formal aition)
  - A is also a primary source of change, a « doer » : it also acts as an efficient aition
- Final aition
  - A form strives toward an end state (e.g. : an object strives to remain in motion once in motion)
  - Aitias of v? No answer for material, efficient, formal
  - The only ENTAILMENT with regard to v is the entailment relationship between v and w
  - The final aition of v (v strives toward w) : <u>the form conveyed by the phrase « v entails w »</u>
- Rosen conclusion :
  - Answsers to « v entails w » cannot be arrived at in a purely syntaxic world
  - <u>Tries to envision a system beyond the limitations of syntaxic Newtonian mechanics, a system with richer entailment structures</u>

Determinations and causal relations (Bailly & Longo)

- Objective determinations
  - Given by invariants relative to symmetries of the considered theory
- Formal determinations
  - Set of rules and equations relative to the considered system
  - Characterize the objects of the theory because make explicit the properties and accessible states of the considered theory
- Properties
  - Do not change when state change
  - If properties change, the object changes
  - Correspond to invariants of efficient causal reduction
- Efficient cause
  - Any cause that can modify accessible states
- Material cause
  - Associated to a modification of properties of objects
- 2 levels of intelligibility
  - Determinations, symmetries, invariants, objects of the theory
  - Causal relations, symmetry breakings (e.g. : asymmetric reading of F=m.a)

State-based models

Rosen reads Newton – The concept of state

- Searching for recursive chronicles :
  - Tf(n)=f(n+1)
  - (Cf different meaning of recursiveness)
- The central concept of newtonian mechanics is the concept of <u>state</u> (LI, p.90)
  - The <u>structureless</u> (nothing inside) particles of the Principia are essentially the atoms posited by the greek analysts
  - 1st law : a particle can not accelerate in an empty environment
    - x''(t)=0
  - 2nd law : effect of environment forces is proportional to acceleration
    - F=mx''
- The upshot of Newton's second law is to effectively collapse the state of the particle, which is an infinite set of variables, down to only two
    - Position and velocity
    - F(x(t),x'(t))=mx''(t) expresses the effect of the environment
- States (phases) constitute a description of the system. Environment is described in termes of a specific recursion rule
- The essential property of a state is its **recursiveness** (LI, p.105)
  - Ie : you can entail one state from an other by a syntactic relation
  - Dual relation between the state and the dynamic law

View of natural systems since Newton

Rosennean complexity

Simple System	Complex System
Fully predicative	Contains impredicativities
Fully fractionable	Contains non-fractionable aspects
Has a single largest syntactic model	Has no single largest syntactic model
Has no complex models	Has complex and simple models
Has computable models	Has noncomputable and computable models
Has no closed loops of entailment	Has closed loops of entailment
Fully syntactic	Has semantic aspects
Synthesis is the inverse of analysis	Synthesis generally distinct from analysis
Epistemology coincides with ontology	Epistemology generally distinct from ontology

Modeling process – Analytic Vs synthetic

Relating newtonian and relational models – Analytic Vs synthetic

- To do biology we need to interrelate newtonian models and relational models (LI, p.153)
- What does it take to bring two formal systems in congruence? How do we model formal objects by other formal objects?
  - In this case ontological and epistemological problems do not exist. We can see and articulate relations between models, and relations between realizations of models
- Examination of modeling within mathematics reveals 2 quiete different approaches to modeling
  - The « analytic », in the context of natural systems, is tied to the notion of **efficient** cause
  - The « synthetic » is tie do the notion of **material** cause
  - In very special situations they happen to be equivalent. These situations all inherently involve some kind of linearity
  - The difference between the analytic and synthetic approaches is closely allied to the difference between syntactic and semantic
    - Gödel's theorem already says that, in a sens to be defined, direct product is bigger
  - The assumption of coincidence between analysis and synthesis leads to the concept of mechanism
  - Complex means that analysis <> synthesis

Remarks concerning equivalence relations

- Equivalence relation $\mathcal{R}$ on set S gives a **quotient set** $S\backslash\mathcal{R}$
- To find s in S, you can first look for its class, then find s in its class
- Geometrically you can see equivalence classes as a base space, and from each point in this space there is a **fiber** made of all the elements of the corresponding class
- If $f : S \to X$ is a mapping, then f induces the equivalence relation R_f defined by sR_fs' iff $f(s) = f(s')$. Therefore $S\backslash R_f \simeq f(S)$ (isomorphism)
- Let's call $f : S \to X$ an **observable** of S, and $f(S)$ the **spectrum** of this observable
- Refinement : $R_{fg} = R_f \cap R_g$. $S\backslash R_{fg}$ contains more classes, and its classes are smaller
- Being given a family $\{f_1, ..., f_n\}$ of observables you can form equivalence relation $\cap_i R_{f_i}$. If this relation is the equality relation on S it means that we can **separate** the elements of S $(\forall(s, s') \in S^2, \exists i | f_i(s) \neq f_i(s')$

Cartesian products and analytical models

- If any class of R_f intersects any class of R_g and vice-versa then :

$$S\backslash R_{fg} \cong f(S) \times g(S)$$

- In this case we shall say that observables f and g are **non-linked**
- Then we have :

$$s\backslash R_f \overset{\Pi_f}{\leftarrow} S \overset{\Pi_g}{\rightarrow} S\backslash R_g$$

- If we have also $\varphi : U \rightarrow S\backslash R_f$ et $\psi : U \rightarrow S\backslash R_g$ then there is a unique mapping $\theta : U \rightarrow S$, defined by $\theta(u) = (\varphi(u), \psi(u))$,such that the following diagram commutes:

$$\begin{tikzcd}
S\backslash R_f & S \arrow[l, "\Pi_f"'] \arrow[r, "\Pi_g"] & S\backslash R_g \\
 & U \arrow[ul, "\varphi"] \arrow[u, dotted, "\theta"] \arrow[ur, "\psi"'] &
\end{tikzcd}$$

Analytical models

- *In what follows, I shall call any expression of a set S as a cartesian product of quotient sets an* ***analysis*** *of S (LI, p.162)*
- The result of an analysis is to **encode** the points of S into a set of coordinates, which are the values of suitable observables of S
- We can compare analysis if one **refines** the other
- It may be that 2 analyses are **incomparable**
- An analysis is a family of observables (non-linked)

$$
\begin{tikzcd}
 & S\backslash \cap_\alpha R_{f_\alpha} \cong \prod_\alpha f_\alpha(S) \arrow[dl, bend left] \arrow[dl, bend right] \arrow[dd, dotted, bend right, "?"'] \\
S \arrow[ur, bend left] \arrow[ur, bend right] \arrow[dr, bend left] \arrow[dr, bend right] & \\
 & S\backslash \cap_\beta R_{g_\beta} \cong \prod_\beta g_\beta(S) \arrow[ul, bend left] \arrow[ul, bend right] \arrow[uu, dotted, bend right, "?"']
\end{tikzcd}
$$

- The idea of linkage between observables is closely tied to the idea of constraint and recursion
- It pertains to the ranges of observables and to entailment in these ranges
- A set S is a cartesian product if we can find a pair of unlinked observables
- We needed a mathematical metaphor for measurements (observations)
  - We have : (real-valued) mappings, their values and their spectra (LI, p.164)
  - Spectra are labels for equivalence classes of elements indistinguishable to such a mapping or observable
- Any analysis gives us an analytic model
- The totality of analytic models can be identified with :
  - The totality of equivalence relations on a set
  - The totality of sets of of observables (the set of all subsets of H(S,R))
- The category of all analytic models has a lot of internal structure, even if we forget about S
- An analytic model looks like a state space. The observables are state variables
  - Questions about S can be pulled into questions pertaining to the ranges of state variables
  - We already did this in our discussion of recursivity (LI, p.165)
- The essence of analytic models is that every element s of S gets encoded into something
  - It gets encoded into the values of the observables

Analytical modeling : examples

- Shannon
- Q Mechs

Shannon Information theory : an example of analytical modeling

Définition 5.1.1 (Source de données). Soit $(X, \mathcal{B}, \mu, T)$ un système dynamique mesuré et $\mathcal{P}$ une partition de X. $(X, \mathcal{B}, \mu, T, \mathcal{P})$ constitue une *source de données*.

Exemple 5.1.2 (Météo). On considère l'atmosphère en un lieu donné comme constituée de n atomes. L'état microscopique du système est la donnée d'un point dans l'espace des phases X de dimension 6n (chaque atome a 6 degrés de liberté). L'état de chaque atome pris individuellement est inobservable. Les fonctions α, β, γ, donnent pour tout x dans X respectivement indiquent l'état de pluie, la température, et le vent associés au micro-état.

Si $\sigma : X \to S$, où S est un ensemble fini, et σ est mesurable, on appelle noyau de σ la partition de X constituée par les pré-images de σ.

Dans notre exemple, si $\mathcal{A}$, $\mathcal{B}$, $\mathcal{C}$ sont les noyaux de α, β et γ, alors $\mathcal{A} \vee \mathcal{B} \vee \mathcal{C}$ représente l'ensemble des états macroscopiques observables. L'atome $A_2 \cap B_3 \cap C_1$ de cette partition représente un état de pluie, une température de 25° et un vent de 30 km/h.

Si $\mathcal{A}$ est le noyau de la "fonction précipitation", la source $(X, \mathcal{M}, \mu, T, \mathcal{A})$ produit un message qui indique si il pleut ou non. Supposons qu'il faille "deviner" chaque heure s'il pleut ou non : la difficulté moyenne des n premiers coups de l'exercice est mesurée par $(1/n).H(\bigvee_{i=0}^{n-1} T^{-i}\mathcal{A})$. Lorsque n augmente on prend en considération de plus en plus de corrélations entre macro-états de précipitation. L'entropie de la source permet de considérer toutes les corrélations entre l'état présent, et tous les états passés et futurs.

Significance of information in Q theory (1) (Grinbaum 2004)

- System : any thing distinct from an other can be treated as a system
- Information : primary substrate of **interaction**
- Facts : facts that provoke information
- Axiom 1 : there is a maximum amount of information one can extract from a system
- Axiom 2 : One can always get new information on a system
- The getting of information is described by Yes-No answers to the questions asked by the observer. The set of questions is written $W(P) = \{Q_i, i \in I\}$
- Axiom 4 (Logical or) $\forall Q_1, Q_2 \in I, \exists Q_3 \in I | Q_3 = Q_1 \vee Q_2$, with also $Q_1 \vee Q_2 = 1$ iff Q_1 or Q_2 equals 1
- Axiom 5 (logical and) $\forall Q_1, Q_2 \in I, \exists Q_3 \in I | Q_3 = Q_1 \wedge Q_2$, with also $Q_1 \wedge Q_2 = 1$ iff Q_1 et Q_2 equal 1
- $W(P)$ is a lattice
- Axiom 6 : $W(P)$ is complete

Significance of information in Q theory (2) (Grinbaum 2004)

- Implication : Question Q_1 ENTAILS question Q_2 (written $Q_1 \rightarrow Q_2$) iff $\neg((Q_{1|M} = 1) \wedge (Q_{2|M} =)0)$, where M represents a measure
- LQuestions Q_1 et Q_2 are said **orthogonal** if $Q_1 \rightarrow Q_2$
- A question Q_2 is said non relevant relatively to Q_1 if $Q_2 \wedge Q_1^{\perp} \neq 0$
- Proposition : lattice $W(P)$ is orthomodular
- Axiom 7 : @ ! ** –(@@@?? (atchoum)
- Hilbert space construction : Be $W(P)$ the set of questions you can ask to a physical system and V a vector space on $\mathbb{D} = \mathbb{R}, \mathbb{C},$ ou $\mathbb{H}$, such that its lattice of subspaces is isomoprphic to $W(P)$. Then there exists a vector product f on V such that V and f form an hilbert space.
- The method can be applied to classical mechanics. The diffrence in Q Mechs is there are questions that can not be asked simutaneously : the set of questions is not a boolean algebra.

Significance of information in Q Theory :

- It is the state notion that becomes relative
- More precisley, relative to observation means

Synthetical modeling

Synthetical models

- An other idea is to encode S as a direct sum of subspaces
- From the category theory point of view, the 2 notions are dual
- Analysis means breaking into coordinates. Synthesis means sticking pieces together
- Also, the idea is to look at S as an image set rather than a source set of observables
- U and U' being disjoint sets, if $i : U \to S$ and $i' : U' \to S$ are two injective mappings, and $\varphi : U \to X$, $\psi : U' \to X$ are mappings, there is a unique mapping $\theta : S \to X$, defined by $\theta(s) = \varphi(s)$ if $s \in U$ or $\theta(s) = \psi(s)$ si $s \in U'$, such that the diagram commutes :

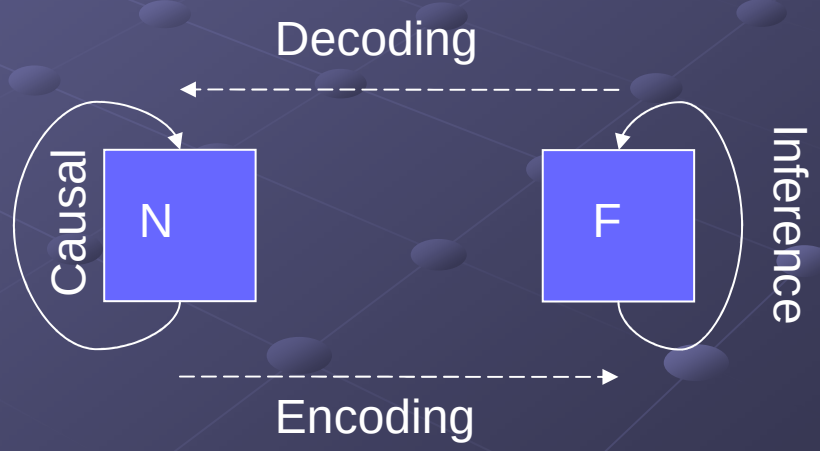
$$\begin{tikzcd}
U \arrow[r, "i"] \arrow[dr, "\varphi"'] & S \arrow[d, dotted, "\theta"] & U' \arrow[l, "i'"'] \arrow[dl, "\psi"] \\
& X &
\end{tikzcd}$$

- For any family of disjoint sets one can produce the sum $S = \sum_\alpha U_\alpha$
- Then consider the analytical models of the U_α and you get the model

$$M(S) = \sum_\alpha M(U_\alpha)$$

- Generally, the encoding of a natural system into formalism is generally unentailed
- The whole purpose of direct sums, in a sense, is precisely to cretae a situation in which encoding is itself entailed (LI, p.170)
  - That is why direct sums have acquired such a dominant place in theoretical science
- The causal status of synthetic states is very different from what we have seen before
  - Being a construct a synthetic state (s in S) is itself an effect, something entailed
  - It is not the answer to a question but something we ask questions about

Modeling relation revisited : symmetries and actions of groups



Symmetries in physics

- S(t0) : state of the system at time t0
- S(t) : state of the system at time t
- T : transformation of the system in state S in a new system in state S'
  - E.g. : expansion of distances, rotation of positions or speeds (constant angle or time dependent), change of charges' sign
- DEF : T is a symetry transformation of the system if, S(t) being a sequence of states, the sequence of states S' is also a possible evolution
  - ie : sequence of states S' describes a system governed by the <u>same evolution laws</u>
  - ie : all possible evolutions are transformed by T in possible evolutions
- Group structure
- Are S and S' sequences necessarily in the same (state) space?
- Something is symetric if after a transformation its apperance is not modified (Weyl)

$$\begin{tikzcd} s(t_0) \arrow[r, "\mathcal{T}"] \arrow[d, squiggly, "\phi"] & s'(t_0) \arrow[d, squiggly, "\phi"] \\ s(t) \arrow[r, "\mathcal{T}"] & s'(t) \end{tikzcd}$$

Non observable	Symétrie par Transformation de:	Loi de conservation
Position spatiale absolue	Translation d'espace	Impulsion P
Temps absolu	Translation dans le temps	Energie E
Orientation spatiale absolue	Rotation	Moment cinétique L
Vitesse, Orientation, Position absolues (Relativité Restreinte)	Transformations du groupe de Poincaré (Lorentz + translations dans l'Espace et le Temps)	Intervalle d'espace temps s^2, Impulsion P, moment cinétique L, Energie E
Orientation, Position, Vitesse et Accélération absolues (Relativité Générale)	- Difféomorphismes (Covariance Générale) - Difféomorphismes infinitésimaux	-Invariants topologiques. - Action (d'Hilbert) champs de gravitation et de matière .
Différence entre particules identiques	Permutation de Particules identiques	Statistique de Fermi Dirac ou de Bose Einstein
Droite (ou gauche) absolue	Changement du vecteur X en le vecteur -X	Parité
Signe absolu de la charge	Changement des particules en leurs antiparticules	Conjugaison de charge
Phase absolue d'un champ de matière chargé	Changement de la phase	Charge électrique
Différence entre mélanges cohérents différents de quarks colorés	Changement de couleur	Générateurs de couleur
Différence entre mélanges cohérents différents de leptons chargés et de neutrinos	Changement d'un lepton en son neutrino	Générateurs d'isospin faible

Renormalization group

- s=(s1,…,sN) : state of N elementary components
- Φ : structure rule, or evolution rule
- Tk : coarse graining operation
- Rk : renormalization operator
- Renormalization group is a symmetry group among others

$$\begin{tikzcd}
\overline{s} \arrow[r, "T_k"] \arrow[d, squiggly, "\phi"] & T_k(\overline{s}) \arrow[d, squiggly, "R_k(\phi)"] \\
\phi(\overline{s}) \arrow[r] & T_k(\phi(\overline{s}))
\end{tikzcd}$$